Automated Verification of English Proficiency Test Scores Using OCR for Graduation Qualification

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***Abstract*—An algorithm is proposed to verify English proficiency scores using OCR for graduation qualification, aiming to reduce errors and workload in the current manual verification process of officer. Score reports in image or PDF format, including TOEIC and KMITL-TEP, are processed using two deep learning-based document classifiers that identify the test type and score type prior to OCR execution. Preprocessing techniques such as noise reduction, contrast enhancement, and skew correction are applied to improve OCR accuracy. Three OCR models—Tesseract, TrOCR, and EasyOCR—are evaluated for text extraction performance. The extracted textual data are then converted into structured JSON, enabling automated rule-based comparison against graduation criteria. Evaluation performance is measured using Accuracy, F1-Score, Character Error Rate (CER), and Word Error Rate (WER). Experimental results show that the proposed system achieves 99% accuracy, demonstrating both high reliability and adaptability to institutional scoring standards. The integration of OCR significantly reduces processing time while maintaining flexibility to accommodate future changes in assessment policies.**

Index Terms*—Optical Character Recognition, English exam score, Tesseract OCR, TrOCR, EasyOCR, Document classification, Score verification.

I. Introduction

In the era of globalization and intense economic competition, English proficiency has become an important skill required for graduates to enter the labor market [1]. In Thailand, the Ministry of Higher Education, Science, Research and Innovation [2] has emphasized the role of English proficiency in the National Qualifications Framework for Higher Education, encouraging institutions to integrate standardized language assessments into academic programs. At the institution selected for this case study, international standardized tests such as TOEIC (Test of English for International Communication) [3], IELTS, TOEFL iBT, and TOEFL ITP, as well as institutional assessments like CU-TEP, TU-GET, and KMITL-TEP are accepted for certifying students' English proficiency. However, due to the sensitive nature of the data and the need to collect information from multiple sources, only TOEIC and KMITL-TEP score reports could be feasibly obtained and were therefore selected as the focus of this study. Nevertheless, the verification of English proficiency test scores at the case-study institution continues to rely largely on manual procedures carried out by registrar staff. This results in several limitations, such as processing delays and the potential for human error. Recent studies in related domains have indicated that manual document verification tasks can lead to error rates of approximately 10–20%, attributable to factors such as fatigue, layout complexity, or cognitive overload [4]. Manual checking also consumes considerable time: clinical form-filling tasks average 6–7 minutes per form, while semi-automated methods reduce this to approximately 3 minutes—a pattern that suggests similar time burdens may apply in administrative document verification [5]. Additional challenges faced by Thai higher education institutions include the inconsistency in document formats, the burden of physical document storage, and the lack of centralized tracking systems [6]. These issues underscore the need for intelligent automation.

Recent advancements in Optical Character Recognition (OCR) have demonstrated strong potential in this domain. For instance, Li and Chen [7] reported that modern transformer-based OCR models such as TrOCR can achieve 95–98% character recognition accuracy on clean, well-structured documents. In addition, Chen et al. [8] proposed a unified OCR benchmarking framework with open-source tools for reproducible evaluation. On the SROIE receipt dataset, their finetuning strategy improved performance from a baseline F1-score of 48.5 and 50.6% CER to 87.8 F1-score and 4.98% CER by splitting images into chunks, while preserving the natural reading order for practical use.

Despite these developments, existing benchmarks and evaluations mostly focus on open-domain document types and do not directly address structured educational documents such as TOEIC or KMITL-TEP score reports. In addition, robust end-to-end systems that integrate document classification, image preprocessing, OCR, and automated score validation in the context of graduation qualification remain underexplored.

This paper presents a novel framework to fill this gap. We propose an end-to-end pipeline leveraging state-of-the-art OCR models—Tesseract, TrOCR, and EasyOCR—with automated preprocessing and document classification to verify English exam scores for graduation eligibility. The system ingests image or PDF score reports, classifies document type using a deep learning classifier, and applies preprocessing steps. (e.g., denoising, deskewing), and subsequently extracts score information through OCR for validation against institutional requirements.

Some previous works related to OCR document are investigated and explained in section II. Details of the proposed algorithm are described in section III, whilst Results of experiments on the actual implementation are shown in section IV. Section V concludes the paper with directions for future research.

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II. Related Works

Optical Character Recognition (OCR) has been extensively researched and applied across various domains, including identity verification, invoice processing, and the digitization of historical documents. Recent advancements in deep learning have significantly improved OCR capabilities, particularly in extracting structured information from documents with complex layouts.

Traditional OCR engines, such as Tesseract, a widely used open-source OCR engine developed by HP and maintained by Google, remains one of the most prominent traditional OCR tools. It performs well on clean, high-contrast documents and supports multiple languages and OCR features, including layout analysis. However, their accuracy tends to deteriorate in the presence of noise, skewed orientations, or diverse typographic styles [9]. To address these limitations, recent approaches have adopted convolutional neural networks (CNNs) and Transformer-based architectures, which provide more robust and context-aware recognition [10].

One notable model is TrOCR, a Transformer-based OCR system introduced by Microsoft, which integrates a Vision Transformer (ViT) encoder with a pretrained language model decoder. Evaluations conducted by Li, Lv, Chen, and Wel [11] demonstrate that TrOCR consistently outperforms traditional OCR engines across a range of degraded document conditions, including blur, low contrast, and watermark interference.

Other models, like EasyOCR, have gained popularity due to their ease of integration and extensibility across multiple languages and scripts. These models combine deep feature extraction networks with sequence decoding mechanisms and have demonstrated favorable results when fine-tuned on task-specific datasets, particularly in non-English and low-resource contexts. Nevertheless, their performance on standardized test documents—which often feature rigid formatting and multi-column layouts—remains an open area of investigation [12].

Moreover, integrating preprocessing stages before OCR, such as denoising, deskewing, and background removal, has been shown to significantly improve overall system performance. In [13], [14] reported that preprocessing can reduce error rates by up to 5ncfc% in noisy or low-quality document images. Complementary research also shows that deep learning–based image rectification methods can further enhance OCR performance, particularly in complex document layouts.

Despite these advances, relatively few studies have specifically investigated OCR applications for educational documents—such as English proficiency test reports (e.g., TOEIC or institutional-specific assessments). These documents often feature tabular layouts and varying field positions, requiring OCR systems to handle both content recognition and structural layout extraction with high precision. To address this gap, this study conducts a comparative evaluation of contemporary OCR models and integrates them into a fully automated workflow for English test score validation in the context of graduation eligibility.

III. Methodology

Our proposed method aims to automate the verification of English exam scores for graduation qualification, as illustrated in Fig. 1. The framework consists of five major stages: image processing, document classification, layout correction and background removal, OCR-based data extraction, and data correction and structured information extraction. These stages are designed to work sequentially to transform raw image or PDF inputs into reliable, structured data that can be validated against institutional graduation criteria. A detailed explanation of each stage is provided below.

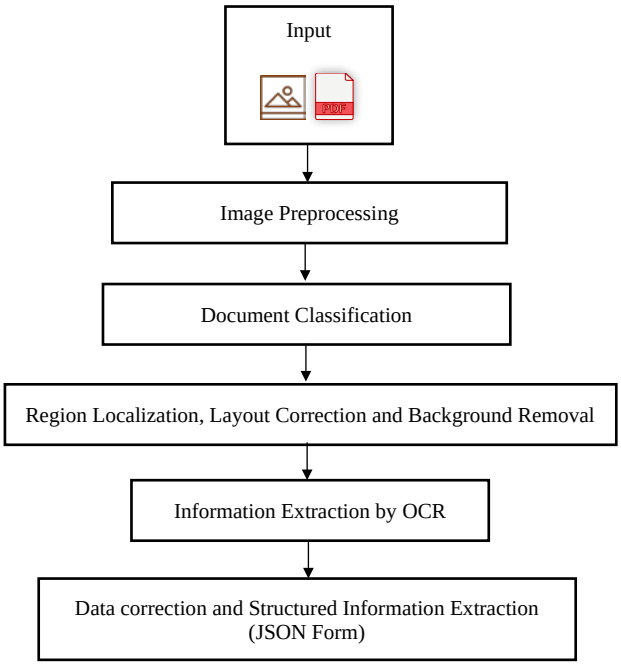

Fig. 1. An overall framework for verifying English exam scores system

The proposed methodology begins with an image preprocessing stage, wherein all input documents—regardless of whether they originate in PDF or image format—are first converted into standardized image files. This step involves essential enhancement operations such as noise reduction, brightness normalization, and skew correction. These automate processes are vital for improving image clarity and ensuring alignment, which significantly impacts the accuracy of subsequent classification and text recognition tasks, as illustrated in Figure 2. The figure shows bounding box predictions for key data classes in a TOEIC score report. In Figure 2(a), where the input image has been preprocessed and properly deskewed, the model is able to detect and localize relevant fields accurately. In contrast, Figure 2(b) demonstrates the consequences of omitting preprocessing: when the document is misaligned, the data classification model fails to correctly capture the intended regions, resulting in incomplete or misplaced bounding boxes. This comparison underscores the critical importance of image preprocessing in enhancing downstream classification performance.

Following preprocessing, the system performs document classification using a deep learning model trained to distinguish between TOEIC and KMITL-TEP score reports. This classification is performed using a retrained Yolov11 model, selected for its ability to handle variations in layout and document format. Accurate classification is essential as the subsequent processing logic, particularly in score field detection and background removal, is tailored to the document type.

(a) (b)

Fig. 2. Bounding box predictions for TOEIC score report components in (a) a deskewed image and (b) a misaligned image. The absence of image preprocessing in (b) leads to inaccurate detection and mislocalization of key fields, highlighting the necessity of proper alignment prior to classification.

Once classified, documents undergo layout correction and background removal. For TOEIC score reports, which typically feature complex, non-standardized layouts, the system further performs layout analysis and component-wise classification. Using additional object detection models—Yolov8s, Yolov8n, and Yolov11n—the system localizes key score components, including: Name, Test date, Valid until, Listening Score, Reading Score, Speaking Score, Writing Score.

Due to layout inconsistencies across different testing centers, robust detection of field positions is essential to ensure the reliability of the extraction process. Figure 3 presents examples of such layout variations and the corresponding bounding box predictions for each score component. In Figure 3(a), the TOEIC score report includes the Listening and Reading sections located in specific regions. In contrast, Figure 3(b) shows another score report where the same physical locations correspond to the Speaking and Writing data classes, but with subtle shifts in alignment and structure. These examples highlight the need for adaptable layout handling mechanisms in the classification and extraction pipeline. In the case of KMITL-TEP documents, the system performs spatial normalization by aligning each key information field— namely, Name, Application No., Date of test Administration, KMITL-TEP Level, as well as individual scores for Grammar, Reading, Listening, and Speaking—according to a predefined system template. This alignment ensures that all critical data classes are mapped to fixed positions, which facilitates consistent processing across varying document versions, as shown in Figure 3(c). Subsequently, the bounding boxes corresponding to each predicted class— specific to each document type— are used to isolate and remove background content. These normalized regions serve as clean and well-structured inputs for OCR-based text extraction, thereby enhancing recognition accuracy and ensuring consistency across varying document layouts.

(a) (b) (c)

Fig. 3. Layout Variability Across TOEIC and KMITL-TEP Score Reports. Examples of layout differences in TOEIC score reports from different testing centers are shown in (a) and (b), where the positions of key fields such as Listening and Reading vary slightly. (c) illustrates a KMITL-TEP score report, which follows a distinct layout and requires alignment to a predefined system template for consistent processing.

The cleaned and corrected documents are then processed by an OCR engine— such as Tesseract, EasyOCR, or TrOCR— which extracts key information including name, application number, test date, and individual or total scores. Despite the use of advanced OCR models, recognition errors may still occur, particularly in structurally important or numerically sensitive fields. For example, characters with similar visual appearance may be misinterpreted (e.g., '8' recognized as 'B', or '1' as 'I'), and extracted date formats are often incomplete or non-standard, requiring character-level parsing and reformatting to match expected formats.

To address these issues, a post-OCR correction module is employed. This module combines rule-based logic with heuristic validation techniques to refine and validate the extracted data, correcting common OCR misinterpretations and ensuring structural consistency across key fields. The rule-based logic component applies predefined constraints to critical fields. For example, the application number is expected to consist of exactly eight digits; test scores must be numeric values within a valid range (e.g., 0–990); and dates must conform to recognizable formats such as DD/MM/YYYY or YYYY-MM-DD. If the OCR output deviates from these expected patterns, the system automatically reformats or corrects the data to match the defined rules. In parallel, the heuristic validation component leverages domain-informed approximations to detect anomalies that may not be captured by strict rules. For instance, if the total score is numerically lower than the sum of its component scores (e.g., Grammar and Reading), the system infers that a recognition error may have occurred. Similarly, if a candidate's name field contains only digits or special characters, it is flagged as likely erroneous. The system also verifies temporal logic: the test date must not be later than the current date at the time of verification, and if an expiration date is specified, it must logically occur after the test date. These heuristics help reinforce the reliability of the final output by identifying and correcting contextually implausible results.

Finally, the corrected data are transformed into a structured output— typically in JSON format— facilitating downstream processes such as automated verification of graduation qualification criteria. This structured representation allows seamless integration with institutional systems, supporting an automated yet reliable evaluation workflow that significantly reduces manual effort while maintaining accuracy and traceability.

IV. Experiments and Results

Experiments were conducted using real-world English proficiency score reports, comprising both TOEIC and KMITL-TEP documents, to assess the effectiveness of the proposed OCR-based verification system. The TOEIC reports were either scanned or captured using digital cameras, while the KMITL-TEP reports were generated directly from the institutional scoring system, as examinees typically receive their results via email in digital format. All processing was performed on a standard desktop computer equipped with an Intel Core i7-8750 CPU and 16 GB of RAM, which provided sufficient computational resources for executing document classification, OCR-based data extraction, and automated verification processes.

A. Dataset

To evaluate the performance of the proposed system, a custom dataset consisting of 253 English proficiency score reports was constructed. The dataset comprises two types of documents: 133 TOEIC score reports obtained from the course Machine Learning at the University of Information Technology, Vietnam National University, Ho Chi Minh City [15], and 120 KMITL-TEP score reports collected from the KMITL-TEP Center, King Mongkut's Institute of Technology Ladkrabang. Each document was manually annotated with bounding boxes corresponding to predefined data fields, including candidate name, application number, test date, validity date, and scores across all relevant skill areas (i.e., Grammar, Reading, Listening, Speaking, and Total score). The annotation process was conducted manually and subsequently verified by domain experts to ensure the accuracy of the ground truth labels.

To enhance model robustness and simulate real-world variability, data augmentation techniques were applied to the original images. These included the addition of Gaussian noise, brightness and contrast adjustments, and random rotations within the range of ±45 degrees. The augmented dataset aimed to expose the models to common distortions encountered in scanned, the institutional scoring system or photographed documents, improving generalization and error tolerance.

Following this, the dataset was expanded to include a total of 798 TOEIC score reports and 756 KMITL-TEP reports. The combined dataset was then partitioned into training (75%), validation (14%), and test (11%) subsets, ensuring a balanced distribution of document types and layout variations across each set.

B. Evaluation Metrics

To evaluate the effectiveness of the OCR-based extraction pipeline, four performance metrics were employed: Accuracy, F1-score, Character Error Rate (CER), and Word Error Rate (WER). These metrics were computed on two versions of the dataset to assess the impact of layout correction and alignment on OCR performance. The first version, raw, unaligned documents, comprised inputs that had undergone only basic noise reduction—reflecting typical conditions of user-scanned or photographed documents, which often exhibit skewed orientation and inconsistent alignment. The second version, preprocessed documents, consisted of the same inputs after applying a full preprocessing pipeline, including deskewing, alignment correction, and noise reduction. By comparing OCR performance across these two conditions, we were able to quantify the contribution of preprocessing techniques to improving recognition accuracy and enhancing the overall reliability of data extraction.

The documents were sourced from multiple test centers to reflect real-world layout variations, particularly in TOEIC reports, which often exhibit non-standardized structures. Although the KMITL-TEP documents generally follow a consistent format, they often contain full-page watermark text, which can introduce noise and pose additional challenges for text detection and recognition. Both image-based and PDF-based inputs were included in the dataset to assess the robustness of the system across diverse document formats. All documents were then processed through a series of image preprocessing steps, aimed at enhancing visual clarity, reducing background interference, and improving the accuracy of subsequent text localization and recognition tasks.

Separate models were trained and fine-tuned for each stage of the processing pipeline. For document classification, the YOLOv11n model was employed, demonstrating efficient performance in distinguishing between TOEIC and KMITL-TEP documents. In the field localization stage for TOEIC documents, a comparative study was conducted using retrained versions of YOLOv8s, YOLOv8n, and YOLOv11n. The objective was to determine the most effective model architecture for detecting key information fields under varying document layout conditions. KMITL-TEP documents, the evaluation involved two distinct approaches: one utilizing predefined templates to specify the expected positions of information fields, and the other operating without templates. These methods aimed to segment and classify data fields prior to applying OCR, thereby improving the accuracy and consistency of text extraction.

For OCR-based text extraction, three widely used engines— Tesseract, EasyOCR, and TrOCR— were evaluated independently to determine their relative effectiveness across document types and formats. Subsequently, the system incorporated a post-OCR correction module, which played a crucial role in improving recognition accuracy and enforcing structural consistency in the extracted data.

C. Result

As previously discussed, image preprocessing constitutes a critical component of the document analysis pipeline. In addition to standard enhancements— such as brightness and contrast adjustments or noise reduction— document orientation correction plays a particularly vital role. Ensuring that documents are properly aligned not only enhances the visual clarity and readability of textual content but also substantially influences the accuracy of both document classification and subsequent information extraction.

For TOEIC score reports, which often feature non-standardized and variable layout structures, rotation correction facilitates more accurate identification of key information fields prior to OCR. The impact of this step is demonstrated in Table 1, where performance improvements are observed across multiple combinations of field localization models and OCR engines.

In contrast, KMITL-TEP documents, while maintaining a relatively consistent format, rely heavily on alignment with a predefined template for reliable field classification. Without proper orientation, template matching becomes infeasible, leading to errors in field detection and OCR processing. The significance of these preprocessing procedures is further validated by the experimental outcomes presented in Table 2, highlighting the role of orientation correction in enabling precise template-based field segmentation and improving text recognition performance.

TABLE I. OCR PERFORMANCE WITH AND WITHOUT ROTATION (TOEIC)

Image Preprocessing (Rotation Correction)	*Field Localization Model*	*OCR Engine*	*Average Accuracy*	*Average F1-score*	*Average CER*	*Average WER*
With Rotation Correction	Yolov 8s	TrOCR	0.9898	0.9907	0.0076	0.0567
		EasyOCR	0.8561	0.8555	0.0795	0.2114
		Tesseract	0.9363	0.9365	0.0515	0.1427
	Yolov 8n	TrOCR	0.9930	0.9938	0.0064	0.0517
		EasyOCR	0.8594	0.8590	0.0745	0.2196
		Tesseract	0.9406	0.9412	0.0481	0.1568
	Yolov 11n	TrOCR	**0.9937**	**0.9946**	**0.0064**	**0.0483**
		EasyOCR	0.8580	0.8573	0.0795	0.2089
		Tesseract	0.9368	0.9376	0.0527	0.1569
Without Rotation Correction	Yolov 8s	TrOCR	0.3574	0.3487	0.6711	0.8371
		EasyOCR	0.2647	0.2655	0.6950	0.8363
		Tesseract	0.1726	0.1671	0.8258	0.9905
	Yolov 8n	TrOCR	0.3926	0.3816	0.6337	0.8113
		EasyOCR	0.2571	0.2568	0.7003	0.8392
		Tesseract	0.1579	0.1539	0.8405	0.9932
	Yolov 11n	TrOCR	0.2108	0.2061	0.7871	0.9255
		EasyOCR	0.1529	0.1532	0.8179	0.9418
		Tesseract	0.1213	0.1187	0.8855	0.9934

Table 1 summarizes the evaluation of various combinations of field localization models (Yolov8s, Yolov8n, Yolov11n) and OCR engines (TrOCR, EasyOCR, Tesseract) under two preprocessing conditions: with and without rotation correction. The goal was to assess how document alignment affects information extraction performance on TOEIC documents.

Results show that applying rotation correction consistently improves performance across all models and OCR engines. Notably, the combination of Yolov11n and TrOCR with rotation correction achieved the highest average accuracy (0.9937) and average F1-score (0.9946), with notably low average CER of 0.0064 and average WER of 0.0483. These findings highlight the importance of rotation correction as a key preprocessing step for robust document analysis in practical settings.

TABLE II. ROLE OF DOCUMENT ALIGNMENT IN TEMPLATE-BASED FIELD CLASSIFICATION (KMITL-TEP)

Preprocessing Approach	*OCR Engine*	*Average Accuracy*	*Average F1-score*	*Average CER*	*Average WER*
With Template Matching	TrOCR	0.9963	0.9967	0.0018	0.0037
	EasyOCR	0.9975	0.9979	0.0016	0.0025
	Tesseract	**0.9977**	**0.9982**	**0.0017**	**0.0023**
Without Template Matching	TrOCR	0	0	100	100
	EasyOCR	0.7717	0.8630	0.2271	0.2283
	Tesseract	0.8180	0.8689	0.1522	0.1820

Table 2 presents the evaluation results comparing two approaches for KMITL-TEP documents: (1) matching templates to identify field classes prior to OCR, and (2) directly applying OCR without template-based classification. The results clearly demonstrate that applying template-based field classification before OCR significantly improves extraction accuracy across all engines. For example, TrOCR combined with template matching achieved the highest overall performance, with an accuracy of 0.9963, F1-score of 0.9967, and minimal error rates (CER: 0.0018, WER: 0.0037). In contrast, skipping the template step led to drastic performance drops, with TrOCR yielding zero accuracy and 100% error rates, indicating complete failure in field recognition without alignment.

Similarly, both EasyOCR and Tesseract exhibited notable declines in all metrics when template matching was omitted, particularly with WER and CER increasing by more than 80-fold. These findings confirm that document alignment and template-based field classification are essential for reliable text extraction in KMITL-TEP documents, where fixed layouts and embedded watermarking require precise preprocessing.

V. CONCLUSIONS

Our proposed method aims to automate the verification of English examination scores for graduation qualification. The framework comprises five major stages: image preprocessing, document classification, layout correction and background removal, OCR-based data extraction, and data correction with structured information formatting. These stages are executed sequentially to convert raw image or PDF inputs into accurate and structured data suitable for validation against institutional graduation requirements. Following document classification, the system proceeds with layout correction and background removal to isolate essential content regions and eliminate visual artifacts—such as watermarks, background noise, or misaligned templates—that may compromise OCR accuracy. This preprocessing step ensures that the input fed into the OCR engine is clean, properly aligned, and optimized for reliable text recognition. In addition, a data correction module is employed to refine extracted text by correcting common OCR misinterpretations and enforcing format consistency. The cleaned and validated data is then transformed into structured JSON format.

This study is limited by its relatively small dataset (253 documents) and coverage of only two document types (TOEIC and KMITL-TEP), which may affect generalizability. Future work will expand the dataset to include additional formats and acquisition conditions, and incorporate benchmarking against commercial OCR solutions once institutional data-sharing constraints are resolved. Further efforts will also address computational performance, scalability under high-volume workloads, and analysis of failure cases, while integrating user feedback and human-in-the-loop strategies to enhance accuracy and operational readiness.

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